

1. The logical binary operation NAND, denoted $|$, is defined by

$$p | q \equiv \sim (p \wedge q).$$

- (a) **(2 marks)** Write down the truth table for $\sim (p | q)$. Show all working.
 (b) **(3 marks)** Prove, using truth tables or otherwise, that $(p | p) \equiv (\sim p)$. Show all working.
 (c) **(3 marks)** Prove, using truth tables or otherwise, that $((p | p) | (q | q)) \equiv p \vee q$. Show all working.
- SOL a)



p	q	$p q$	$\sim (p q)$
T	T	F	T
T	F	T	F
F	T	T	F
F	F	T	F

- SOL b)



p	$(p p)$	$\sim p$
T	F	F
F	T	T

Since the last 2 columns are the same the propositions are equivalent.
 OR

$$\begin{aligned} p | p &\equiv \sim (p \wedge p) \\ &\equiv \sim p \end{aligned}$$


- SOL c)

$$\begin{aligned} ((p | p) | (q | q)) &\equiv (\sim p) | (\sim q) \\ &\equiv \sim (\sim p \wedge \sim q) \\ &\equiv \sim (\sim (p \vee q)) \\ &\equiv p \vee q. \end{aligned}$$


OR



p	q	$((p p) (q q))$	$p \vee q$
T	T	F	T
T	F	T	T
F	T	T	T
F	F	T	F

The fact that column 4 and column 6 are the same shows the propositions are equivalent.

2. (a) Consider the statement

There exists a real number x such that for all real numbers y , $xy = 1$.

- (i) (2 marks) Write the negation of this statement.
 (ii) (2 marks) State which of the original statement or its negation is true. Justify your answer, showing all working.

(b) (4 marks) Consider the statement

For all integers a and b , if $(ab)^2$ is odd then a is odd and b is odd.

Use either a direct proof, a proof by contraposition or a proof by contradiction to prove this statement is true. Show all working.

- SOL (a) i) The negation is

For all real numbers x , there exists a real number y so that $xy \neq 1$.

- ii) The negation is true: For all real numbers x , choose $y = 0$. Then $xy = 0 \neq 1$.

- SOL (b)

The statement is equivalent to

$$\forall a, b \in \mathbb{Z}, (ab)^2 \text{ is not odd and, } a \text{ is odd or } b \text{ is odd.}$$

This statement is TRUE.

The contrapositive statement is

$$\forall a, b \in \mathbb{Z}, \text{ if } a \text{ is even or } b \text{ is even, then } (ab)^2 \text{ is even.}$$

Assume $a = 2p$ and $b = 2q$ where $p, q \in \mathbb{Z}$. Then

$$ab = 2p \cdot 2q = 2(2pq) = 2x, \text{ where } x \in \mathbb{Z}$$

and

$$(ab)^2 = 2(2x^2), \text{ where } 2x^2 \in \mathbb{Z}.$$

Thus $(ab)^2$ is even as required.

for logical setting out
 & statements like
 $p \in \mathbb{Z}$

3. (a) (2 marks) Let $A = \{0, 1, 2, 3, 4\}$, $B = \{3, 4, 5, 6\}$ and $C = \{1, 2, 4, 5, 7\}$ and let

$$R = ((A - B) \cap (A - C)) \cup (B \cap C \cap A).$$

List the elements of the set R .

- (b) (2 marks) Let $X = \mathcal{P}(\emptyset)$ and $Y = \mathcal{P}(\mathcal{P}(\emptyset))$. List the elements of each of the following sets.

- (i) X
- (ii) Y
- (iii) $X \cap Y$
- (iv) $X \cup Y$

- SOL a) Note $A - B = \{0, 1\}$ $A - C = \{0, 3\}$ and $(A - B) \cap (A - C) = \{0\}$, so a list of the elements are 0, 4. ①
- SOL b) The sets and the lists of elements are below.

$X = \{\emptyset\}$. list- $\emptyset$

$Y = \{\emptyset, \{\emptyset\}\}$. list- $\emptyset, \{\emptyset\}$

$X \cap Y = \{\emptyset\} = X$. list- $\emptyset$

$X \cup Y = \{\emptyset, \{\emptyset\}\} = Y$. list- $\emptyset, \{\emptyset\}$

$\frac{1}{2}$ mark for each underlined bit

4. (5 marks) Prove that $|\mathbb{Z}^+ \times \mathbb{Z}^+| = |\mathbb{Z}^+|$.

- SOL A solution that does not use a diagram will invoke the Schröder-Bernstein theorem.

Consider the function $f : \mathbb{Z}^+ \rightarrow \mathbb{Z}^+ \times \mathbb{Z}^+$ with $f(x) = (x, 0)$. ½

This function is injective, because if $f(x_1) = f(x_2)$, then $(x_1, 0) = (x_2, 0)$, so $x_1 = x_2$. ½

Therefore $|\mathbb{Z}^+ \times \mathbb{Z}^+| \geq |\mathbb{Z}^+|$. !

On the other hand, consider the function $g : \mathbb{Z}^+ \times \mathbb{Z}^+ \rightarrow \mathbb{Z}^+$ with $g((x, y)) = 2^x 3^y$. ½

Then if $g((x_1, y_1)) = g((x_2, y_2))$, then $2^{x_1} 3^{y_1} = 2^{x_2} 3^{y_2}$. ½

By the unique prime factorisation theorem, we have $x_1 = x_2$ and $y_1 = y_2$ so $(x_1, y_1) = (x_2, y_2)$.

Therefore, g is injective, and $|\mathbb{Z}^+ \times \mathbb{Z}^+| \leq |\mathbb{Z}^+|$. !

The S-B theorem then implies that $|\mathbb{Z}^+ \times \mathbb{Z}^+| = |\mathbb{Z}^+|$. !

5. (a) **(4 marks)** Let a be an integer such that the remainder is 1 when a is divided by 4. Prove that when a^4 is divided by 16 the remainder is also 1.

(b) **(4 marks)** Use Euclid's Algorithm to verify that $\gcd(2000, 1713) = 1$. Clearly show all steps in the execution of the algorithm.

- SOL a) Let $a = 4k + 1$, for $k \in \mathbb{Z}$. !

Thus $a^2 = 16k^2 + 8k + 1 = 8k(2k + 1) + 1$. Let $k' = k(2k + 1) \in \mathbb{Z}$ and so $a^2 = 8(k') + 1$, where $k' \in \mathbb{Z}$. !

Then $a^4 = (a^2)^2 = (8(k') + 1)^2 = 64(k')^2 + 16k' + 1 = 16k'(4k' + 1) + 1 = 16k''$, where $k'' = k'(4k' + 1) \in \mathbb{Z}$. !

- SOL b)

$$2000 = 1 \times 1713 + 287$$

$$1713 = 5 \times 287 + 278$$

$$287 = 1 \times 278 + 9$$

$$278 = 30 \times 9 + 8$$

$$9 = 1 \times 8 + 1$$

$$8 = 8 \times 1 + 0$$

(2)

(2)

Hence $\gcd(2000, 1713) = 1$.

6. (8 marks) Use mathematical induction to prove that the following statement is true.

For all integers $n \geq 0$, $3^n < (n+3)!$

- SOL Let $P(n)$ be the statement that $3^n < (n+3)!$.

Basis Step Prove $P(0)$ is true. $3^0 = 1$ and $(0+3)! = 3! = 6$. $1 < 6$ so $P(0)$ is true. 2

Inductive Step Assume $P(k)$ is true, so $3^k < (k+3)!$. 1

Prove $P(k+1)$ is true, so prove $3^{k+1} < (k+1+3)!$.

$$\begin{aligned}
 LHS = 3^{k+1} &= 3 \times 3^k && \text{1} \\
 &< 3 \times (k+3)! && \text{1} \\
 &< (k+4) \times (k+3)! \text{ since } k \geq 0 \text{ and so } 3 < k+4 && \text{1} \\
 &= (k+4)! = (k+1+3)! = RHS \text{ as required.} && \text{1}
 \end{aligned}$$

So $P(k+1)$ is true and by PMI $P(n)$ is true for all $n \geq 0$. 1

if setting out is not logical
students should not receive
full marks for

7. Consider the function $f : \mathbb{Q} \cap [1, \infty) \rightarrow \mathbb{Q} \cap [2, \infty)$ given by

$$f(x) = (x - 1)^2 + 2 \text{ for all } x \in \mathbb{Q} \cap [1, \infty).$$

(a) **(4 marks)** Prove that the function f is injective.

(b) **(3 marks)** Prove that the function f is not surjective.

- SOL a) Suppose that there are $x_1, x_2 \in \mathbb{Q} \cap [1, \infty)$ such that $f(x_1) = f(x_2)$.
Then $(x_1 - 1)^2 + 2 = (x_2 - 1)^2 + 2$, so that $(x_1 - 1)^2 = (x_2 - 1)^2$.
Since $x_1 - 1$ and $x_2 - 1$ are both non-negative, we conclude that $x_1 - 1 = x_2 - 1$, so that $x_1 = x_2$.
Thus, f is injective.
- SOL b) This question relies on the observation that the square root of a rational number is not always rational.
For example, when $y = 4$, and we try to find x so that $f(x) = 4$, we need to find an $x \in \mathbb{Q}$ so that $(x - 1)^2 = 2$, but this is impossible since $x - 1$ must be rational.

8. Consider the relation ρ on $\mathbb{Z}$ given by

$$a\rho b \text{ if and only if } (a-b)(a+b) = 4k \text{ for some } k \in \mathbb{Z}.$$

- (a) **(6 marks)** Prove that ρ is an equivalence relation.
 (b) **(2 marks)** Determine the number of equivalence classes of ρ . Justify your answer.

- SOL a) Note $(a-b)(a+b) = a^2 - b^2$. So we prove

$$a\rho b \text{ if and only if } a^2 - b^2 = 4k, \text{ for some } k \in \mathbb{Z}.$$

For any $x \in \mathbb{Z}$, $x^2 - x^2 = 0 = 4 \cdot 0$, so $x\rho x$. Therefore, ρ is reflexive. ①

For any $x, y \in \mathbb{Z}$, if $x\rho y$, then $x^2 - y^2 = 4k$ for some $k \in \mathbb{Z}$, so $y^2 - x^2 = 4(-k)$, with $-k \in \mathbb{Z}$, so $y\rho x$. Thus, ρ is symmetric. ①

Finally, if $x\rho y$ and $y\rho z$ for $x, y, z \in \mathbb{Z}$, then $x^2 - y^2 = 4k_1$, and $y^2 - z^2 = 4k_2$ for some integers k_1, k_2 . ①

Then $x^2 - z^2 = x^2 - y^2 + y^2 - z^2 = 4(k_1 + k_2)$, with $k_1 + k_2 \in \mathbb{Z}$, so $x\rho z$. Therefore, ρ is transitive. ①

Since ρ is reflexive, symmetric and transitive, ρ is an equivalence relation. ①

- SOL (b) By the quotient remainder theorem, every integer x can be written as $x = 4k + r$, for some $r \in \{0, 1, 2, 3\}$. ½

But $(4k+1)^2 - (4k'+3)^2 = 4k'' - 8 = 4(k'' - 2)$, also $(4k)^2 - (4k'+2)^2 = 4k'' - 4 = 4(k'' - 1)$ ½
 so the equivalence classes are $[0], [1]$. ½

Therefore, there are two distinct equivalence classes. ½

9. (a) **(1 marks)** Determine whether the groups $(\mathbb{Z}, +)$ and $(\mathbb{R} - \{0\}, \cdot)$ are isomorphic. Justify your answer.
- (b) **(3 marks)** Determine whether the groups $(\mathbb{Z}_4, +)$, $(\mathbb{Z}_5 - \{[0]\}, \cdot)$ are isomorphic. Justify your answer.
- (c) **(4 marks)** Find a subset S of $\mathbb{Z}_{10}$ such that $(S, \cdot)$ is a group of order 4 (where $\cdot$ denotes multiplication modulo 10). Justify your answer.

• SOL a) Not isomorphic because $\mathbb{Z}$ and $\mathbb{R} \setminus \{0\}$ have different cardinalities, so there cannot be a bijection between them.

• SOL (b) Isomorphic because

$(\mathbb{Z}_4, +)$ is cyclic, generated by 1 ($1^2 = 2, 1^3 = 3, 1^4 = 1^0 = 0$).

Also, $(\mathbb{Z}_5 - \{[0]\}, \cdot)$ is cyclic, generated by 3 ($3^2 = 4, 3^3 = 2, 3^4 = 1 = 3^0$).

Therefore the required isomorphism $\phi : \mathbb{Z}_4 \rightarrow \mathbb{Z}_5 - \{[0]\}$ can be found with $\phi(0) = 1, \phi(1) = 3, \phi(2) = 4, \phi(3) = 2$.

• SOL (c). Clearly such a set needs to contain 1, being the identity.

It also needs to contain 9, being the only integer between 0 and 9 which squares to something that is 1 mod 10.

Now, since there are no other such elements that are their own inverses, this particular group must be cyclic, so the other elements need to be 3 and 7 which both square to 9 mod 10.

So the group is $\{1, 3, 7, 9\}$. I think it is the only one (but this is not necessary for the solution).

OR

They could just guess the set and then show that it is a group.

10. For this question, you may leave your answers as expressions involving factorials.

- (a) **(2 marks)** How many distinct arrangements can be made of the letters of the word WOOLLOONGABBA?
- (b) **(3 marks)** If the letters of the word WOOLLOONGABBA are permuted at random, what is the probability that we get the same word back?
- (c) **(3 marks)** How many distinct arrangements are there of the letters of the word WOOLLOONGABBA such that exactly one of the following is true?
- The two As are not next to each other.
 - The two Bs are not next to each other.
 - The two Ls are not next to each other.

- SOL(a) 13 letters, O occurs four, L occurs twice, A occurs twice, B occurs twice, so there are

$$\frac{13!}{4!2!2!2!} = \frac{13!}{4!.8} = 13 \times 12 \times 11 \times \cdots \times 9 \times 7 \times 6 \times 5.$$

- SOL(b) There are 13! possible ways of permuting the letters, all equally likely. There are only $4!2!2!2!$ ways of getting the same word back (because of possible random permutations repeated letters). So the probability is

$$\frac{4!2!2!2!}{13!} = \frac{4!.8}{13!} \quad \text{for prob.}$$

- SOL(c) There are three possibilities:
- As and Bs are adjacent, Ls are not;
 - Bs and Ls are adjacent, As are not; or
 - As and Ls are adjacent, Bs are not.

Suppose that the As and Bs occur as adjacent pairs.

Then since they are attached, there are essentially only 11 letters left, so $11!$ permutations. But because of the repeat letters (O and L in this case), there are only $\frac{11!}{4!2!}$ distinct arrangements. On the other hand, if the As, Bs and Ls all appear as adjacent pairs, then there are $\frac{10!}{4!}$ distinct arrangements.

Therefore, there are $\frac{11!}{4!2!} - \frac{10!}{4!}$ ways of having As and Bs adjacent, but Ls not adjacent.

Therefore, since the three possibilities have no overlap, the answer is

$$3 \cdot \left(\frac{11!}{4!2!} - \frac{10!}{4!} \right)$$

11. (a) **(4 marks)** Let T be a tree on p vertices where $p \geq 11$, T has precisely six vertices of degree 1 and precisely four vertices of degree 3. Determine the degree of each of the remaining $p - 10$ vertices of T .
- (b) **(4 marks)** Let d be a positive integer and let T be a tree such that T has a vertex of degree d . Prove that T has at least d vertices of degree 1.

• SOL a)

T is a tree with p vertices, and so $p - 1$ edges.

Then

$$2(p - 1) = \sum_{i=1}^p d(v_i) = 6 \times 1 + 4 \times 3 + \sum_{i=1}^{p-10} d(v_i).$$

So

$$2(p - 10) = \sum_{i=1}^{p-10} d(v_i)$$

and the average degree of any vertex is 2. Since there can be no more vertices of degree less than 2 all remaining vertices have degree 2.

- SOL b) Assume T has p vertices and so it has $p - 1$ edges. Let x represent the number of vertices of degree 1 and we have

$$2(p - 1) = \sum_{i=1}^p d(v_i) = 1 \times d + x \times 1 + \sum_{i=1}^{p-1-x} d(v_i),$$

where $2 \leq d(v_i) \leq d$. So

$$\begin{aligned} d + x + 2(p - 1 - x) &\leq 2p - 2 \\ d &\leq x, \end{aligned}$$

and there are at least d vertices of degree 1.

$\frac{1}{2}$